# Problem K. Magic

Input file: standard input
Output file: standard output

**Warning: Unusual memory limit!**

You are given a sequence $a_0, \ldots, a_{2n}$. Initially, all numbers are zero.

There are $n$ operations. The $i$-th operation is represented by two integers $l_i, r_i$ ($1 \le l_i < r_i \le 2n, 1 \le i \le n$), which assigns $i$ to $a_{l_i}, \ldots, a_{r_i-1}$. It is guaranteed that all the $2n$ integers, $l_1, l_2, \ldots, l_n, r_1, r_2, \ldots, r_n$, are distinct.

You need to perform each operation exactly once, in arbitrary order.

You want to maximize the number of $i$ ($0 \le i < 2n$) such that $a_i \neq a_{i+1}$ after all $n$ operations. Output the maximum number.

## Input

The first line contains an integer $n$ ($1 \le n \le 5 \times 10^3$).

The $i$-th line of the next $n$ lines contains a pair of integers $l_i, r_i$ ($1 \le l_i < r_i \le 2n$). It is guaranteed that all the $2n$ integers, $l_1, l_2, \ldots, l_n, r_1, r_2, \ldots, r_n$, are distinct.

## Output

Output one integer representing the answer in one line.

## Example

| standard input | standard output |
|---|---|
| 5<br>2 3<br>6 7<br>1 9<br>5 10<br>4 8 | 9 |